Class 2

Consonants

PARAMETERS FOR DESCRIBING CONSONANTS

1 Voicing and aspiration

- **1 Voicing and aspiration** (= what the larynx is doing)
  - Voiced sound = vocal folds vibrating (e.g., /z/)
  - Voiceless sound = vocal folds **not** vibrating (e.g., /s/)
  - Aspiration = vocal folds held apart long enough to allow an extra puff of air to exit the mouth
- **2 Place of articulation** (which articulators are moving, and where)
- **3 Manner of articulation** (what sort of constriction and how much)

2&3 Place of articulation

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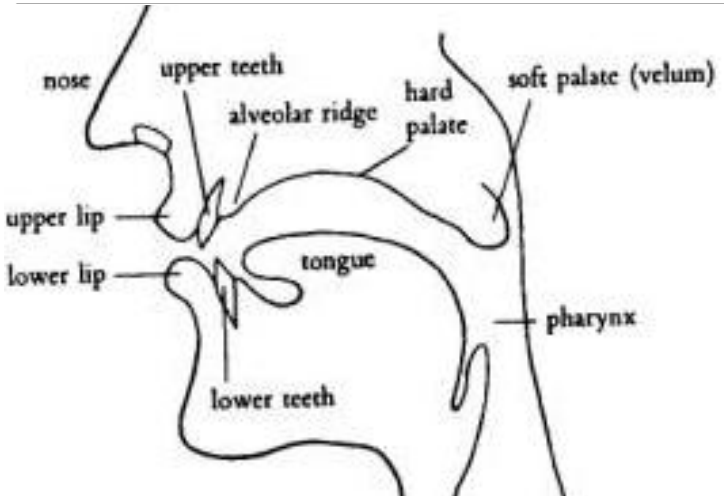
Manner and Place of articulation

CONSONANTS (PULMONIC) © 2005 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k g	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			



Labial	Coronal	Dorsal	Radical	Laryngeal
bilabial	(inter)dental	velar	Pharyngeal	Glottal
labiodental	alveolar	uvular		
	postalveolar			
	retroflex			
	palatal			

3 Manner of articulation

- **Oral stop/plosive**: complete closure, pressure builds up
  - Active and passive articulators are brought together to make a complete closure so that airflow out of the mouth is cut off

[p]

- **Nasal stop**: open velum, resonance in nasal cavity
  - The velum is open and air flows freely out the nose, but there's complete closure somewhere else

[m]

- **Fricative**: incomplete closure, turbulent air
  - Articulators are brought close together but not closed completely; the stream of air forced between them becomes turbulent and noisy

[v]

- **Affricate**: stop + fricative
  - A sequence of stop plus fricative combined to form a single sound

[dʒ]

- **Approximant**: narrowed vocal tract, but no frication
  - The active articulator moves to narrow the vocal tract, but not so much that fricative noise is created
  - Subtypes:
    - glides
    - liquids (laterals & rhotics)

[j]

3 Manner of articulation

Approximants

- **Glides** - [w] ‘well’ and [j] ‘you’
  - Vowel-like quality
- **Laterals** - [l]
  - Air flows over sides of tongue
  - American [l] is alveolar; what about laterals in other languages (e.g. Spanish, Italian?)
- **Rhotics** - [ɹ]
  - The American English ‘r’ is upside-down in the IPA because it is a retroflex consonant
  - Rhotics vary widely across world’s languages - examples?

Manner and Place of articulation

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Nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
Trill	ʙ		r						ʀ		
Tap or Flap		ⱱ	ɾ			ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ	ɹ			ɻ	j	ɰ			
Lateral approximant			l			ɭ	ʎ	ʟ			

We classify consonants by place and manner of articulation

(a) voiced or voiceless?
(b) place? (bilabial, alveolar, etc.)
(c) manner? (stop, fricative, affricate, etc.)

EXERCISES

Consonant Practice

Exercise 1: Make the sound

- (a) Voiced alveolar (oral) stop
- (b) Voiceless labiodental (oral) fricative
- (c) Voiced bilabial nasal stop

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Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Consonant Practice

Exercise 2: Make the sound and describe the parameters:

- (a) [p]
- (b) [ɹ]
- (c) [g]

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Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Exercise 3:

- voiced velar nasal stop [ŋ]

  _____

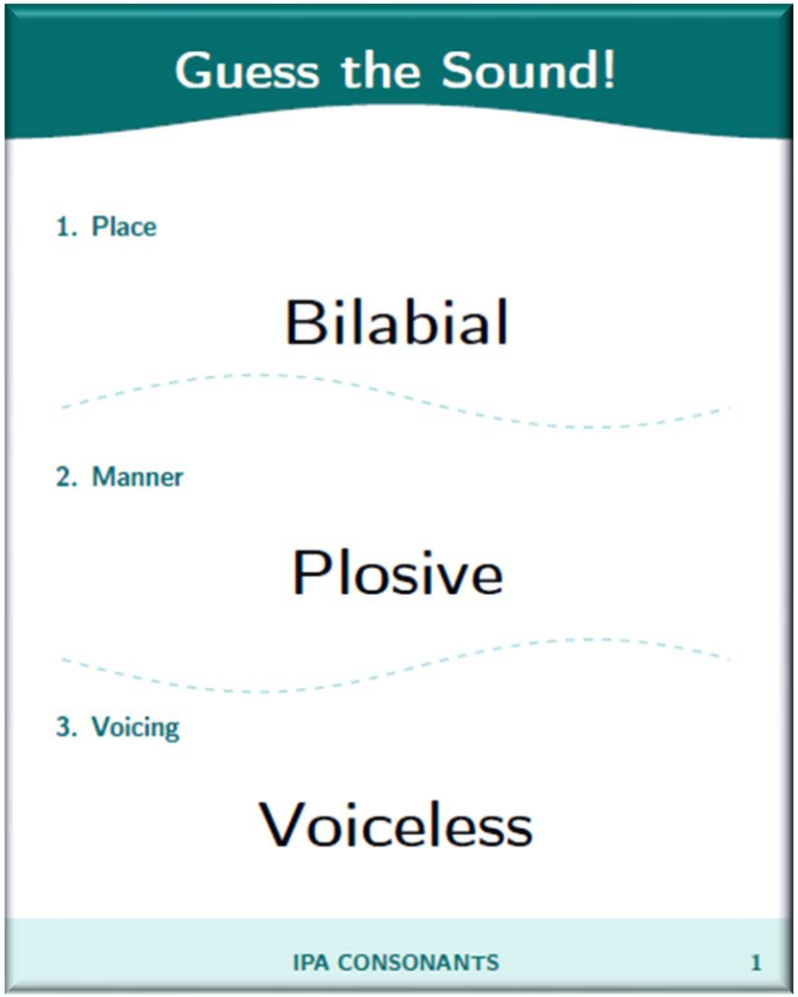
- voiceless oral glottal fricative [h]

  _____

- voiceless oral alveopalatal affricate [ʧ]

  _____

Exercise 4:


Guess the Sound!
1. Place
Bilabial
2. Manner
Plosive
3. Voicing
Voiceless
IPA CONSONANTS
1

Exercise 4:

Guess the Sound!

1. Place

Bilabial

2. Manner

Plosive

3. Voicing

Voiceless

IPA CONSONANTS 1

Exercise 4:

Guess the Sound!

1. Place

Bilabial

2. Manner

Plosive

3. Voicing

Voiceless

IPA CONSONANTS 1

[p]

CONSONANT CHART

	Bilab.	L-D.	Dent.	Alv.	P-A.	Pal.	Vel.	Glo.
Plosive	p b			t d			k g	ʔ
Nasal	m			n			ŋ	
Tap/Flap				ɾ				
Fricat.		f v	θ ð	s z	ʃ ʒ			h
Approx.	w			ɹ		j		
Lat. Appr.				l				

IPA 10 questions

Goal: Guess the hidden sound using only Yes/No questions!

How to Play:

1. **Draw** one card. Keep it a secret!
2. **Ask** "Yes/No" questions to guess the card.
3. **Think** about Place, Manner, and Voicing.
4. Whoever shouts the correct IPA sound first gets the card!

🗣️ **Useful Phrases:**

YES:

"Is it voiced / voiceless?"
"Is it an alveolar sound?"

NO: "Tell us about the manner of articulation of that sound."

IPA Karuta

Goal: Be the first to grab the correct sound!

How to Play:

1. **Spread** the cards face-up on your desk.
2. **Hands on your head!** (Wait for the teacher).
3. **Listen** carefully to the sound or word.
4. **Slap** the card with the matching features!

   (e.g., Hear /m/ ➔ Grab "Bilabial / Nasal / Voiced")

⚠️ **Rule:**
False start? You must sit out for one turn!